

Real Estate Math in Michigan

STUDY SHEET

Basic Math Concepts

Area

- Square, rectangle, or parallelogram = length x width
- Triangle = $(0.5 \times \text{base}) \times \text{height}$

Price per Square Foot

- Price per square foot = sales price $\div$ area

Loan-to-Value Ratio

- The typical housing debt-to-income ratio used by lenders is 25 to 28% (based on gross income)
- The typical total debt to income ratio used by lenders is 33 to 36% (based on gross income)
- Loan amount $\div$ home value = loan-to-value ratio

Loan Origination Fee

- One point is equal to 1%
- Loan amount x loan origination percentage = loan origination fee

Loan Discount Fee

- One point is equal to 1% of the loan value
- Loan amount x number of points (%) = loan discount fee

Taxes and Prorations in Michigan

General Property Tax

- Tax assessed value x tax rate = property tax
- When working with ad valorem real estate tax, use the tax-assessed value

Michigan Transfer Tax

- Total transfer tax is \$8.60 per \$1,000: The county in which the property is located receives \$1.10 per \$1,000 and the state receives \$7.50 per \$1,000.
- Customarily, this tax is paid by the seller, but this is negotiable.

Calculating Commission

- Sales price x commission rate = commission • Sales price x percent to seller = net dollars to seller

Mortgage Transaction Calculations

Gross Income

Gross income is what a person makes. If you subtract taxes and other expenses from the gross income, you're left with net income. This is what the person "brings home."

Maximum House Payment

- Gross income x housing debt-to-income ratio = maximum house payment

Interest

- Loan balance x interest rate = annual interest amount

Calculations for Valuations

Gross Income

- Gross income is the income received before expenses are deducted. Once the expenses are deducted, this is the net operating income.

Calculating Depreciation

- $\text{Original value} \times \text{percentage of value not lost} = \text{present depreciated value}$

Gross Rent Multiplier and Gross Income Multiplier

- Gross rent multiplier (GRM) is the relationship of a property's sale price to its gross income.
 - $\text{Sales price} \div \text{monthly gross rent} = \text{gross rent multiplier}$
- Gross income multiplier (GIM) is used to determine value for five or more units.
 - $\text{Sales price} \div \text{annual gross income} = \text{gross income multiplier}$